GeoMarket Terminal

A zero-licence-cost desktop terminal for geopolitical and market intelligence

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Abstract

Professional market terminals bundle prices, news, macro data and analytics in one screen, but a single seat costs around $32,000 a year [Costbench(2026)]. I designed and built GeoMarket Terminal on my own, as a desktop application that gives an individual a comparable *workflow* at no licence cost. It combines market prices, a news feed from more than 70 sources, official macro statistics, sanctions lists, live vessel and aircraft positions, and a country-level risk score. All of this sits on an interactive 2D map and 3D globe, with an AI analyst that summarises the context on request. In this note I describe what the application does, the constraints I set myself, what I changed in version 1.0.6 after reviewing my own code, and what is still missing. I do not cover implementation details.

1 Why I built it

Markets react to events that happen far from the trading screen: a tanker diverted around a chokepoint, new sanctions on an exporter, GPS interference near a conflict zone, a surprise in an economic release. Professional desks follow these signals through expensive terminals and specialised data vendors; I was stitching together a dozen browser tabs. With GeoMarket Terminal I put these sources side by side, on a map, with a common timeline, so that I can answer the question *"what is happening, where, and which assets does it touch?"* from one place.

2 Design constraints

I built the application around four constraints, which I set before writing the code:

1. **Zero licence cost.** Every data layer runs on public or free-tier sources. Where a source is unreliable, the terminal keeps a ranked chain of alternatives and falls back automatically.
2. **Local-first and lightweight.** The application runs on an ordinary laptop with no external database server. I wrote down performance budgets and develop against them: start-up under 30 seconds, under 500 MB of memory at rest and under 1.5 GB under heavy load, on CPU-only machines with 4–8 GB of RAM.
3. **Transparent sources.** A source-health page shows which feeds are live, stale or failing, so I can tell a quiet market from a broken feed.
4. **AI as an assistant, not an oracle.** The AI analyst works only from data stored in the terminal. It can switch between a local model and cloud providers, and it keeps a record of the claims it makes so that I can check them later.

3 What the application does

Area	What the user sees
Overview	World map or 3D globe with country risk rankings, geopolitical events, bilateral tensions and a heat map
Maritime and air	Live vessel and aircraft positions, chokepoint traffic statistics, GPS-jamming zones, and a notice when the live feed is down and cached data is shown
News feed	Articles from more than 70 international sources, grouped into stories and tagged by country, sentiment and likely market impact
Macro and calendar	Official macro indicators and an economic calendar showing past and upcoming releases
Sanctions	Official sanctions lists, searchable and linked to the countries on the map
Charts and options	Price charts with event markers; options volatility surface and unusual activity
Correlations	Statistical links between unusual event activity in a country and the next-day return of an asset, with significance tests
Portfolio	Positions, P&L and a trading journal
Strategy Lab	Event-driven strategy ideas, each tested against random entry dates and on a separate verification period
Alerts	Risk spikes and custom alerts, with optional delivery to a messaging bot that answers only authorised chats
AI analyst	Daily brief, calendar outlook and free-form questions answered with the data held by the terminal

Table 1: Functional areas of GeoMarket Terminal 1.0.6.

4 Architecture (overview)

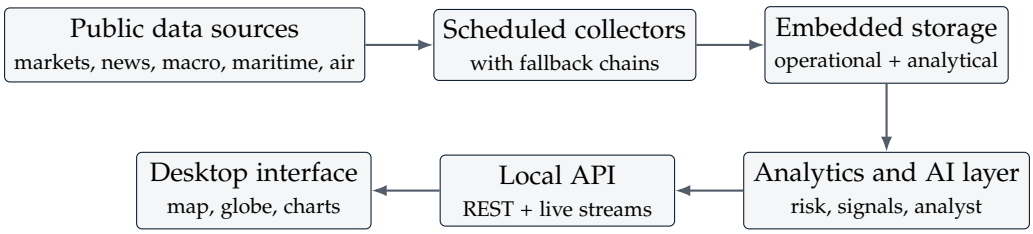


Figure 1: Data flow, from public sources to the desktop interface. Everything runs on the user's machine.

I wrote the backend in Python; it exposes a local web API. The interface is a lightweight web front end inside a native desktop window, and a Windows installer packages everything. Data is collected on a schedule, stored in embedded databases and analytical files, and served to the interface through both request–response endpoints and live streams.

5 What I changed in version 1.0.6

Before publishing, I reviewed my own code with the same standards I apply to my trading research. The most important changes:

- **Strategy Lab.** The old evaluation counted the same price move several times when signals overlapped, and its Sharpe ratio treated overlapping five-day returns as daily ones. Now each signal opens at most one trade per asset, entered on the next trading day and net of 10 basis points of costs. The Sharpe ratio uses the actual number of trades per year, each strategy is compared with 200 runs on random dates, and the metrics are reported separately for the first 60% and the last 40% of the period.
- **Correlations.** This page used to apply fixed rules. It now measures the Pearson correlation between a country's unusual event activity (a z-score against its own previous 60 days) and the next-day return of each asset, over one year, with t-statistics, p-values and a Bonferroni correction for the many pairs tested.
- **Country risk score.** Half of the event-frequency component now measures the change against

the same country's 30-day average, so that heavily covered countries no longer look risky only because they are in the news every day.

- **Robustness and security.** The installer ships with an empty configuration, and the build stops if it finds anything that looks like a credential. Certificate checks are on by default, the alert bot answers only authorised chats, and errors that used to be silently ignored are now logged. I added automated tests for the backtest, the statistics and the risk score.

6 Current limitations

- **Free data has limits.** Some feeds are delayed, rate-limited or patchy in coverage. The terminal is a research and situational-awareness tool, not an execution-grade data feed.
- **The risk score is still a heuristic.** It is now normalised by country, but it has not yet been validated against subsequent market moves.
- **Correlations need history.** With one year of daily data and a correction for multiple tests, few pairs pass the significance threshold. I treat a significant pair as a hypothesis to test, not as a trading signal.
- **Single user, local only.** The application has no user accounts and is meant to run on one machine.

7 Next steps

I plan to add a historical analytics layer over the stored price files, show which source served each data point, split the two largest modules into smaller ones, and record a short public demo.

References

[Costbench(2026)] Costbench. *Bloomberg Terminal pricing 2026* (third-party estimate: $31,980 per year for a single terminal). `https://costbench.com/software/financial-data-terminals/bloomberg-terminal/`, accessed 14 October 2026.

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